

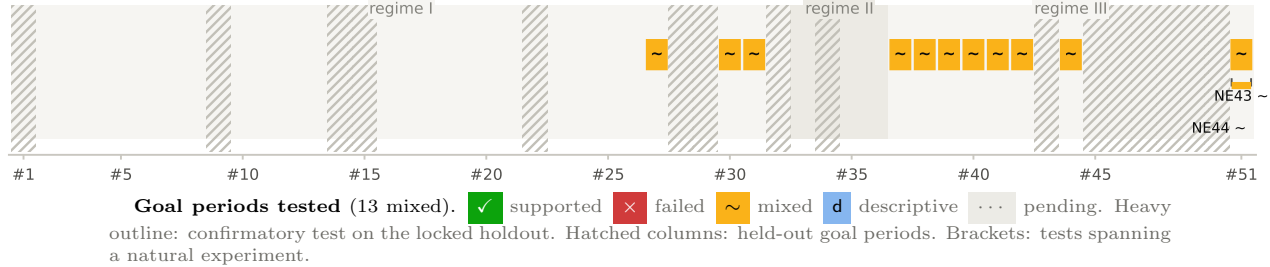
H16: Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 48% Faithfulness 1.5/5 Usefulness 2.5/5

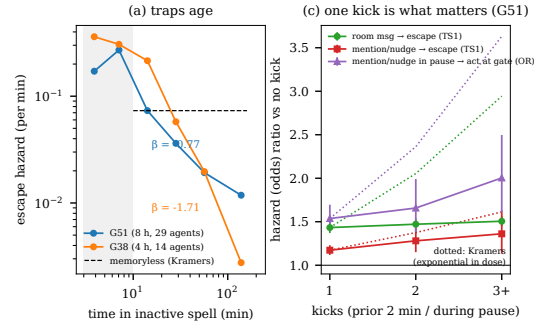
Bottom line. Agent traps are real and individual and they age by themselves (unchanged when the nudger stops), not by Kramers escape; kicks act as saturating triggers at timer gates, and real failure loops age only in the long #51 window.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Trap types (inactive spells, timer-pause chains, error loops, repeated-command loops) were tested in 11 non-holdout goal periods. Kramers predicts memoryless escape; instead traps *age*: within agent the escape hazard falls roughly as $t^{-0.8}$ (G51 $\beta = -0.77$; G38 -1.71), and each re-pause lowers the odds of acting at the next timer expiry (G51 -0.38). One directed message during a pause raises those odds $\times 1.5$ – 2.9 (5/7 periods); more add little. Round 1b (improved data) counted real failures instead of stderr, used the leading-@ nudge target and added Jev blocked and loop spells. “Error loops age” was largely a stderr artifact (rows -51% ; aging only in #51, -0.57). Jev blocked spells age in #51 and #27; kick results are unchanged; with the nudger off (NE43) aging is unchanged. $\beta J_0 = 0.09$ – 0.41 , no bistability. Holdout not run.



Observables. (a) Escape hazard from inactive spells vs. time in the spell (robust spells, ignoring single stray actions) in the two best-powered periods. Kramers predicts the flat dashed line once the first minutes (grey) have passed; instead the hazard falls, so traps age. (b) G51 hazard and odds ratios by number of kicks. Dotted lines extrapolate Kramers (exponential in dose) from one kick. The observed curves flatten: one message is what matters.

PHYSICAL MODEL

Models 01/02 (mean field) plus Kramers escape. Agent activity x moves in a double well $U(x)$ (idle and active basins) at temperature T . A trap is a stay in the idle basin, escaped at rate k over barrier ΔU (attempt rate ν). A kick (a message reaching the agent) adds force f . Swarm: Curie–Weiss coupling J_0 , field h , active fraction m . Fitted: hazard h_{it} with agent effects α_i , aging slope β , dose effects γ_n .

$$k = \nu e^{-\Delta U/T}, \quad G(x) = -\ln P(x) = U(x)/T + \text{const}$$

$$k(n) = k_0 e^{n f/T} \text{ (Kramers)} \quad \text{vs} \quad k(n) = k_0(1 + c n) \text{ (triggers)}$$

$$\ln[-\ln(1 - h_{it})] = \alpha_i + \beta \ln t + \sum_n \gamma_n \mathbb{K}[\text{dose}_{it} = n]$$

$$f(m) = -\frac{1}{2} J_0 m^2 - h m - T s(m); \text{ bistable iff } \beta J_0(1 - m^2) > 1$$

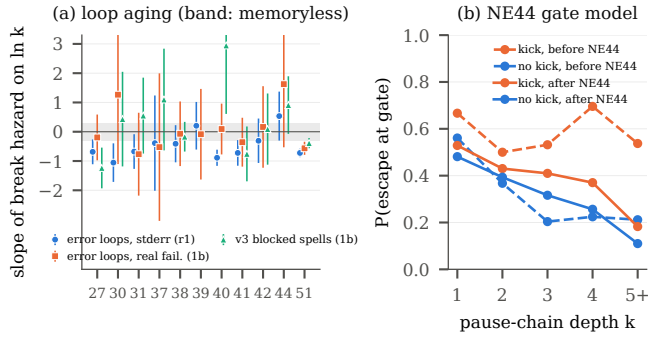
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Modestly useful, mostly as a correction. A stuck agent gets stickier the longer it sits, and switching the nudger off does not change that, so the deepening is the agent’s own loop: intervene early. The lever that works is one directed message landing during a pause, read at the next timer expiry; repeats and room chatter add little, and under today’s 5-min pause default a kick barely helps an agent deep in a pause chain. Messages never break failure loops; those need tooling or a reset. Real failure loops are rarer and shorter than stderr suggested. The physics added little: “Kramers” failed and “aging plus saturating triggers” is ordinary survival analysis.

H16-metastable-traps-kramers · generated 2026-10-04 00:27 by infra/summaries/build_summaries.py · card hypotheses/H16-metastable-traps-kramers/README.md · ratings: H16 re-evaluation agent RE-B1 (suggested; round-1 values were coordinator-calibrated)

RESULTS VS PREDICTIONS

prediction (round 1b)	observed	verdict
real-failure loops (majority)	age 3/10 point, 0/10 CI; G51 failed -0.57	
messages do not break them	9/9 periods	holds
Jev blocked spells (G51, G38, $\geq$ half)	age G51 -0.39, G27 -1.24; G38 failed flat; timer-like in 3	
Jev loop spells (G51)	age -0.46 [-0.55, -0.18]	partial
gate kick OR (leading @)	≥ 1.5 5/7 (1.5-2.9)	holds
room kicks above null	6/8	holds
NE44 kick $\times \ln k$	0 +0.57 before, +0.06 after then < 0	failed
NE44 declared pauses shrink	90 s $\rightarrow$ 180 s	failed
NE43 aging without TS1r	-0.59 $\rightarrow$ -0.53; gates -0.31 $\rightarrow$ -0.26	holds
NE43 gate escape drops	+0.006 [-0.09, 0.10]	failed



Second figure. (a) Loop aging slope per period: error loops on stderr (round 1) vs real failures (round 1b), and Jev blocked spells; the gray band is memoryless. Only #51 keeps a clear negative slope on real failures. (b) Gate escape by pause-chain depth: before NE44 a directed kick rescues agents at any depth; after it, deep chains barely respond.

GOAL PERIODS IN DETAIL

G27 ~ mixed
G30 ~ mixed
G31 ~ mixed
G37 ~ mixed
G38 ~ mixed
G39 ~ mixed
G40 ~ mixed
G41 ~ mixed
G42 ~ mixed
G44 ~ mixed
G51 ~ mixed
NE43 ~ mixed
NE44 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The NE44 contrast is between periods (the change sits inside the locked holdout window), confounded with 4 h vs 8 h days, roster and goals. Jev **p_blocked** means “waiting on others” in regime I and “failing or looping” in regime III, and window spells are short in 4-h periods. Kick labels are temporal coincidences. Several 3–5-day periods have unstable day-bootstrap CIs (Wald reported alongside). The landscape and swarm blocks were not re-run (inputs unchanged).

ROUND 2

Pólya-urn test of self-reinforcement from the context ledger (own-output share vs aging); content-level spirals on the holdout (#45).

- **Essence.** Traps deepen by self-reinforcement as the context fills with the agent’s own output.
- **H16-R1** Pólya urn: $P(\text{repeat}) \propto$ the share of the context that is own repeats.
- **H16-R2** One directed message works because it dilutes the urn; repeated messages add little once the context is diluted.
- **H16-R3** Trap depth is a model trait: some models loop more.